RIPHAH
INTERNATIONAL UNIVERSITY

BS-AI

LAB TASK 10

Name: Hafiz Abdullah Shakeel

Sap I'D: 72405

Subject: Object Orientated Programming

Teacher: Sir Imran Khan

Task 1:

```
ublic class Lab10 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("Name: Hafiz Abdullah Shakeel");
        System.out.println("SAP ID: 72405");

        System.out.print("Enter first number: ");
        double num1 = sc.nextDouble();

        System.out.print("Enter second number: ");
        double num2 = sc.nextDouble();

        Calculator calc;

        calc = new Addition();
        System.out.println("Addition: " + calc.calculate(num1, num2));

        calc = new Subtraction();
        System.out.println("Subtraction: " + calc.calculate(num1, num2));

        calc = new Multiplication();
        System.out.println("Multiplication: " + calc.calculate(num1, num2));

        calc = new Division();
        System.out.println("Division: " + calc.calculate(num1, num2));


    }
```

Lab10.java

Source History

```java
package lab10;
import java.util.Scanner;

class Calculator {
    public double calculate(double a, double b) {
        return a + b;
    }
}

class Addition extends Calculator {
    public double calculate(double a, double b) {
        return a + b;
    }
}

class Subtraction extends Calculator {
    public double calculate(double a, double b) {
        return a - b;
    }
}

class Multiplication extends Calculator {
    public double calculate(double a, double b) {
        return a * b;
    }
}

class Division extends Calculator {
    public double calculate(double a, double b) {
        if (b != 0) {
            return a / b;
        } else {
            System.out.println("Cannot divide by zero!");
            return 0;
        }
    }
}

public class Lab10 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("Name: Hafiz Abdullah Shakeel");
        System.out.println("SAP ID: 72405");

        System.out.print("Enter first number: ");
```

Output - lab10 (run)

```
run:
```

Task 2:

Lab10.java Lab102.java

Source History

```java
package lab102;
import java.util.Scanner;

class Shape {
    public double area(double r) {
        return 0;
    }
    public double area(double r, double h) {
        return 0;
    }
}
class Circle extends Shape {
    public double area(double r) {
        return Math.PI * r * r;
    }
}
class Cylinder extends Shape {
    public double area(double r, double h) {
        return 2 * Math.PI * r * (r + h);
    }
}
public class Lab102 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Name: Hafiz Abdullah Shakeel");
        System.out.println("SAP ID: 72405");
        System.out.print("Enter radius: ");
        double radius = sc.nextDouble();
        System.out.print("Enter height (for cylinder): ");
        double height = sc.nextDouble();
        Shape shape;
        shape = new Circle();
        System.out.println("Circle Area: " + shape.area(radius));
        shape = new Cylinder();
        System.out.println("Cylinder Surface Area: " + shape.area(radius, height));
        sc.close();
    }
}
```